
Auditing Density Uncertainty in Cryo-EM Beyond a Gaussian Reconstruction Dictionary

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Preliminary experiments; not a completed submission candidate.

Abstract

A confidence interval can be calibrated within a reconstruction dictionary and still miss errors excluded by that dictionary. We investigate this failure in cryogenic electron microscopy (cryo-EM), using conjugate-paired Fourier Gaussians as a computational representation. Our candidate procedure estimates fixed linear density features and audits their bias through the adjoint residual in an explicitly larger voxel space. This separates numerical representation from the density class to which a confidence statement applies. We specialize classical bias-aware inference to the Fourier-slice model, derive explicit bounds for local pose errors, and implement reduced-space and matrix-free solvers with primal-dual diagnostics. Development experiments include repeated bounded-model tests and acquisition geometries from three experimental particle stacks. In fixed-pose tests, dictionary-only intervals can severely undercover an independent map generator; an ambient-space audit restores the assumed-model guarantee. Broader features admit useful intervals, while finer features and restricted viewing angles expose large uncertainty. Density radii, support, Gaussian noise and pose bounds remain explicit assumptions. Comprehensive end-to-end validation, several external baselines and independent review are outstanding; these preliminary results do not establish publication readiness.

1. Introduction

Cryo-EM reconstructs molecular density from noisy projections whose orientations, translations and imaging parameters are imperfectly known. Modern neural and Gaussian

representations offer expressive reconstruction models, but expressiveness alone does not establish which reconstructed features the data support. We focus on a fixed local density average or contrast. This target differs from a conformational population, the uncertainty of an atomic model, or prediction of a noisy held-out particle image.

An uncertainty calculation usually inherits its reconstruction space. If a Gaussian dictionary spans S , a posterior or confidence interval computed in S need not cover density outside S . Improving optimization inside the dictionary does not remove this discrepancy. Independent half reconstructions can also share the same representational restriction. Our working question is whether one can retain Gaussian computation while auditing feature uncertainty in a larger, explicitly declared density class.

We develop three connected components: an adjoint-residual audit that separates fitting and inference spaces; a Fourier-Gaussian treatment of bounded local pose errors, including the density-pose interaction and a uniform nonlinear remainder; and a validation protocol that distinguishes fixed-parameter coverage, Bayesian prior-predictive coverage and experimental-image prediction. We do not claim a new general theory of bias-aware intervals, reduced-space duality or primal-dual optimization. The value of the cryo-EM specialization must be established by its usefulness, assumptions and comparison to existing methods.

2. Related Work and Scientific Scope

Probabilistic reconstruction. Bayesian inference is foundational in cryo-EM (Scheres, 2012). Ullrich et al. (2020) already combine differentiable Fourier-slice reconstruction with variational density uncertainty and discuss the distinction between variance and model bias. Rangan et al. (2024) examine soft modes of the joint volume/alignment likelihood Hessian. Thus neither a probabilistic Fourier representation nor pose-density ambiguity is a new observation. Our candidate emphasizes a finite-sample, bounded-class statement and a check outside the fitting dictionary.

Representations and heterogeneity. CryoDRGN learns a latent-conditioned density family (Zhong et al., 2021a); its latent variation is not automatically calibrated uncertainty in

¹Fourier Cryo Splats research project. Correspondence to: Project repository <<https://github.com/yashpatel5400/fourier-cryo-splats>>.

a homogeneous density. Ab initio neural inference (Zhong et al., 2021b) is a different task from fixed-pose backprojection. Gaussian mixtures and Gaussian splatting are also established (Chen & Ludtke, 2021; Chen & Ling, 2026; Qu et al., 2025; Shekarforoush et al., 2025). We use Fourier kernels as a numerical basis, without treating their existence as the contribution. Continuous-state recovery, conformational populations and pose entanglement are substantial separate problems; the living survey records current RECOVAR, DynaMight, cryoDRGN-AI and CAHRA work with source-level distinctions.

Validation and map confidence. Gold-standard splitting and noise substitution address reconstruction overfitting (Scheres & Chen, 2012; Chen et al., 2013); a shell correlation is not a coverage statement for a local feature. LocScale-2.0 provides calibrated confidence for map enhancement relative to its reference target (Bharadwaj et al., 2026). That target must not be conflated with confidence in an unknown density from raw particles. Map significance relative to background noise, directional resolution, atomic-model reliability and particle-derived confidence likewise answer different questions.

Statistical foundations. Optimal recovery links linear inverse estimation to bias–variance tradeoffs (Donoho, 1994). Honest inference over convex function classes and bias-aware Gaussian critical values are classical (Armstrong & Kolesár, 2018). Our theorems below specialize these principles and identify the assumptions that cryo-EM introduces. The survey and its bibliographic ledger are ongoing; a down-loaded paper is not counted as a completed critical review.

3. Forward Model and Inference Target

With $F(k) = \int \rho(x) e^{-2\pi i k^\top x} dx$, a particle obeys

$$Y_i(q) = C_i(q) e^{-2\pi i q^\top t_i} F(E_i q) + \epsilon_i(q). \quad (1)$$

E_i embeds a detector plane, C_i is its contrast transfer function (CTF), and frequencies and translations use reciprocal units. We realify one representative of each conjugate Fourier pair and whiten the noise. The initial theory conditions on a fixed design and an independent pilot ρ_0 , giving

$$y - \mathcal{A}\rho_0 = \mathcal{A}\delta + Ju + r + \epsilon, \quad \epsilon \sim N(0, I). \quad (2)$$

Here $\delta = \rho - \rho_0$, J is block diagonal over particles, u describes bounded nuisance errors, and r is a bounded remainder. We assume

$$\|\delta\|_H \leq B, \quad \|u_i\| \leq \eta_i, \quad \|r_i\| \leq \gamma_i. \quad (3)$$

H is either a declared Gaussian density subspace or an ambient voxel space with a stated support mask. These

are different classes. A physical density-energy norm includes voxel volume; experiments using an ordinary voxel Euclidean norm explicitly normalize that norm. B , η_i and γ_i are assumptions, not quantities automatically learned from residual fit.

The estimand is $t = \langle \ell, \rho \rangle$, such as a Gaussian-weighted spatial average or the difference of two such averages. The smoothing scale and coordinate frame are specified in physical units. For weights fixed without the inference noise, use

$$\hat{t} = \langle \ell, \rho_0 \rangle + w^\top (y - \mathcal{A}\rho_0). \quad (4)$$

The pilot, dictionary, whitening, target and weights may depend on independent development data. Poses estimated from the inference images violate literal fixed-design independence; supplied consensus poses therefore support only a conditional experimental analysis here.

4. Bias-Aware Confidence and Ambient Auditing

4.1. A conditional certificate

Define $s = \|w\|$ and

$$b_H(w) = B \|\ell - \mathcal{A}^* w\|_H + \sum_i \eta_i \|J_i^\top w_i\| + \sum_i \gamma_i \|w_i\|. \quad (5)$$

Let $q(s, b, \alpha)$ solve

$$\Phi((q - b)/s) - \Phi((-q - b)/s) = 1 - \alpha \quad (6)$$

when $s > 0$, and set $q(0, b, \alpha) = b$.

Theorem 1 (Conditional coverage). *Under Equations (2)–(3), the interval $\hat{t} \pm q(s, b_H(w), \alpha)$ covers t with probability at least $1 - \alpha$, uniformly over the stated class, conditional on the independent design choices.*

Proof. The deterministic estimation error is $\langle \mathcal{A}^* w - \ell, \delta \rangle + \sum_i w_i^\top J_i u_i + \sum_i w_i^\top r_i$. Cauchy–Schwarz bounds its absolute value by b_H . The remaining error is Gaussian with standard deviation s . Coverage of a symmetric interval decreases with the absolute Gaussian mean, so the minimum occurs at mean $\pm b_H$ and equals Equation (6). $\square$

The result is a direct bias-aware inference construction. It is useful only if its assumptions are defensible and its width improves on the no-data interval $\langle \ell, \rho_0 \rangle \pm B \|\ell\|$.

4.2. Why auditing the dictionary matters

Let $S : \mathbb{R}^p \rightarrow H$ have orthonormal columns. A restricted calculation replaces the density term by $B \|S^*(\ell - \mathcal{A}^* w)\|$. Write $h = \ell - \mathcal{A}^* w$. Then

$$\|h\|_H^2 = \|S^* h\|^2 + \|(I - SS^*)h\|_H^2. \quad (7)$$

The second term measures uncertainty discarded by the dictionary. It can be large even when the restricted residual is zero. Evaluating the full adjoint residual repairs this omission for H without estimating a representation-error direction from noisy data. It does not cover density outside the declared ambient grid or support.

Centering in Equation (4) also uses the ambient operator. If a surrogate centers the images instead, the discrepancy $w^\top(\mathcal{A} - \tilde{\mathcal{A}})\rho_0$ requires an additional bound. We do not silently substitute a surrogate forward model into the coverage theorem.

4.3. Optimization and verifiable stopping

We optimize the conservative convex objective

$$\mathcal{F}_H(w) = z_{1-\alpha/2} \|w\| + b_H(w), \quad (8)$$

then report the sharper width from Equation (6). Its dual is

$$\begin{aligned} \max_{v,a,d} \quad & \langle \ell, v \rangle \\ \text{s.t.} \quad & \|v\| \leq B, \quad \|a_i\| \leq \eta_i, \quad \|d_i\| \leq \gamma_i, \\ & \|\mathcal{A}v - Ja - d\| \leq z_{1-\alpha/2}. \end{aligned} \quad (9)$$

Representing each norm by a Euclidean support function proves this dual; continuity and coercivity yield strong duality. A feasible dual iterate supplies a lower bound before convergence. The gap concerns Equation (8), not optimality of the folded-normal width.

For a reduced density space S , a feasible dual vector lifts to Sv in H . Thus a restricted dual lower bound and an ambient primal upper bound bracket the same full-space optimum. We enrich S with the component of $\ell - \mathcal{A}^*w$ orthogonal to it, using only design and target information. Restricted exact optima increase under enrichment; individual ambient primal values need not decrease, so we keep the best feasible iterate. No convergence-rate claim is made for this greedy construction.

Quadratic majorization gives coefficient-space conjugate-gradient solves with particle-block inverses. A classical Chambolle–Pock implementation is an alternative (Chambolle & Pock, 2011). For fixed poses, a full-space matrix-free comparator solves $(\mathcal{A}^*\mathcal{A} + \lambda I)v = \ell$ within majorization. It can outperform enrichment and is included to prevent an unsupported Gaussian speed claim.

5. Fourier Gaussians and Local Pose Errors

We use smooth conjugate pairs

$$\begin{aligned} F_c(k) &= \sum_j c_j g_j(k) + \bar{c}_j g_j(-k), \\ g_j(k) &= e^{-\|k - \mu_j\|^2 / (2\sigma_j^2)}. \end{aligned} \quad (10)$$

This enforces Hermitian symmetry. Hard cutoff kernels from the earlier reconstruction prototype are unsuitable for an unqualified Taylor bound; the uncertainty implementation uses full Gaussians. Their analytic whole-space Gram matrix provides density-energy coordinates $\delta = T\beta$ with $T^*GT = I$, so $\|\beta\| \leq B$ is a density-energy assumption rather than an arbitrary coefficient scale.

Parameterize local rotations by a Lie-algebra vector and translations by detector shifts. Normalize their joint vector so $\|u_i\| \leq 1$, with physical radii incorporated in the operator. With $D_{ia} = \partial A_i / \partial u_{ia}$ at the pilot pose,

$$\begin{aligned} A_i(u_i)(c_0 + \delta) &= A_i(0)(c_0 + \delta) + J_i u_i \\ &\quad + \sum_a u_{ia} D_{ia} \delta + e_i. \end{aligned} \quad (11)$$

If H_i uniformly bounds the pilot’s second derivative along all allowed segments and K_i bounds the corresponding operator derivative, then $\|e_i\| \leq (H_i + BK_i)/2$. Define $T_i(w_i) = [D_{i1}^\top w_i, \dots, D_{iq}^\top w_i]$. The interaction contributes at most

$$B \sum_i \|T_i(w_i)\|_{\text{op}} \leq B \sum_i \|T_i(w_i)\|_F. \quad (12)$$

The implemented Frobenius relaxation is a sum of Euclidean group norms. It ignores some sharing of δ across particles and can be conservative. Charging the entire first-order interaction to an isotropic remainder was substantially worse in development.

The finite-Gaussian pose experiment and the full-voxel fixed-pose audit are currently separate. Combining the ambient audit with useful full-grid nonlinear nuisance bounds remains unfinished. Fixed CTFs and known whitened Gaussian noise are also assumptions; CTF error, window correlations and heterogeneous densities are not covered by assertion.

6. Development Experiments

All results below are exploratory. A confirmatory protocol has not been frozen. Experimental particle pools have already been inspected during the earlier reconstruction study and subsequent development; their partitions must not be described as untouched final tests.

6.1. Data, references and provenance

We retain 8,192 extracted experimental images from each of EMPIAR-10028, 10049 and 10076 (EMPIAR, 2015a;b; 2016). Source exposure groups are reconstructed from deposited metadata. EMPIAR-10076 requires the original Frealign FILM field because its STAR micrograph numbers are sequential particle identifiers. Hash-based whole-group partitions provide pilot, tuning, inference-half and test labels

within this development pool. Consensus poses and CTFs remain conditional inputs.

Deposited maps EMD-2660, EMD-6487 and EMD-8434 define independent voxel signal generators after explicit low-pass resampling. They are simulation truth, not proof of experimental accuracy. Nonuniform Fourier transforms project these maps independently of the Gaussian representation. Native and downloaded-file hashes, source indices, signs, field sizes and software versions are recorded.

6.2. Representation and exact-dictionary controls

A noiseless projection audit with 256 particles per split and a radius-six frequency band found relative held-out errors of approximately 1.8%, 2.7% and 7.7% for the finest tested Gaussian dictionary. These residuals demonstrate why noise-only intervals do not cover representation error.

The bounded-model coverage benchmark has 48 settings: 12 seeds, broad versus preferred orientations, and rotation radii 0.5° or 2° . Each setting uses 96 particles, 40 coefficients, 20 random bounded signals and estimator-specific adversarial controls. We report exact Gaussian noise-marginal coverage and cross-check with 2,000 paired noise draws per signal. Shared designs and paired noise do not constitute independent voxel trials.

The structured certificate’s smallest tested coverage was 0.991 at nominal 0.95; omitting the quadratic remainder gave 0.932. Median width relative to the no-data bound was 0.557 for broad views at 0.5° , and approximately one for preferred views. This is a usefulness limitation, not a calibration success to conceal with pooled averages.

6.3. Ambient-space audit

The fixed-pose audit uses 128 real acquisition geometries per dataset, a 24^3 voxel grid, a spherical support of radius 0.35 times the field, and prescribed signal-to-noise power ratio 0.1. A pilot derived from real particles and each reference generator are normalized to unit voxel Euclidean norm. We set $B = 2$ before seeing coverage; the triangle inequality covers all such normalized generator/pilot pairs. This simulation convention does not estimate the density radius of experimental data.

The initial dictionary has 33 non-null Gaussian modes. Two targets, a central average and an axial contrast, use Gaussian averaging widths 0.03 and 0.07 times the field. Restricted intervals, the same weights with a full-grid audit, enriched weights, and the ambient matrix-free solution use identical estimands. The larger-grid model is deliberately able to express errors outside the initial dictionary.

Table 1 reports median widths across the two targets. The full-space solver reached its 0.5% objective-gap tolerance

in 12 of 12 cases, taking 2.4–11.1 seconds per target on the Mac. The enrichment solver reached the same outer tolerance in 8 of 12 cases within 100 rounds; nonconverged results retain valid bounds but do not establish optimality. The solvers optimize the conservative sum-width, so their sharper reported widths need not be ordered exactly by their objective gaps.

Dictionary-only intervals can be narrow because they exclude admissible density errors. For the EMPIAR-10028 central broad average, analytic coverage on the independent voxel-map generator is approximately 4.2×10^{-5} . The same image weights, with the ambient bias bound, cover at the prescribed level over the ambient ball. On each estimator’s constructed ambient bias boundary, the audited intervals attain nominal 0.95 Gaussian coverage up to numerical tolerance. These boundary calculations check the theorem and implementation; they are not independent evidence of practical class calibration. Broader targets have substantially smaller relative widths than fine targets. The prescribed norm radius and fixed-pose assumption are essential to interpreting that result.

6.4. Baselines and numerical checks

Implemented matched mathematical baselines include full Gaussian posterior covariance, diagonal variational covariance, regularized sampling covariance, joint Gaussian pose covariance, identifiable unregularized estimation, fixed-pose bias-aware intervals, coarse and structured nuisance bounds, and the no-data interval. These are not executions of every external paper’s original code. Full Gaussian posteriors pass their exactly matched linear prior-predictive identity at 0.95. Narrow priors can nevertheless fail bounded fixed-signal tests; wider priors and unregularized estimation can produce very wide intervals.

Independent conic optimization checks the majorization, primal-dual and ambient solvers on small instances. Derivative finite differences, nonlinear remainder tests, direct Fourier sums, adjoint identities and independent Gaussian-integral checks validate implementation components. The current suite has 30 passing tests. Tight FINUFFT tolerances are numerical checks, not interval-arithmetic guarantees on roundoff.

7. Limitations and Work Remaining

The current evidence does not establish an end-to-end uncertainty method with estimated poses, noise law and density class. Local pose radii and the total density bound are assumptions. Selecting them from the same inference residuals would require new analysis; a narrow residual does not rule out an almost invisible density error. Learned support can exclude true structure. Gauge and hand ambiguity must

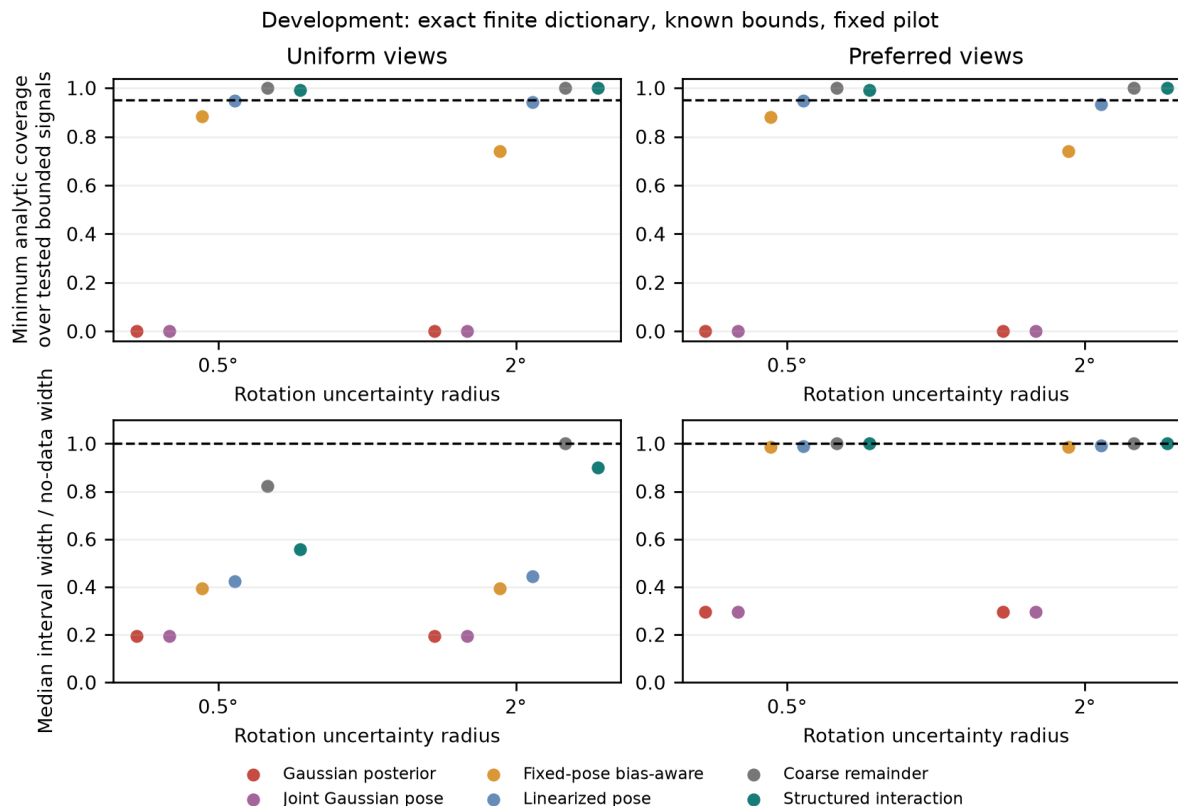


Figure 1. Exact-dictionary development tests with known bounds and an independent fixed pilot. Minimum coverage is over the finite tested set, including adversarial signals; it is not an empirical proof of the worst case. Widths count each independent design once. Gaussian posterior methods separately pass their correctly matched prior-predictive coverage check at 0.95. Their fixed-parameter boundary failures do not contradict Bayesian prior-average calibration.

be fixed or included in the target class.

The next validation must freeze the method before fresh simulations and additional source groups, include higher bandwidth and grid refinement, compare bootstrap and actual neural reconstruction pipelines, test correlated noise, CTF errors, heterogeneity and support violations, and quantify coverage–width–power tradeoffs. Simultaneous maps need multiplicity correction. Real-data prediction and stability cannot supply unavailable density-coverage labels. Comprehensive literature screening and independent review of a complete candidate remain outstanding.

Reproducibility and Impact

The active research branch is uncertainty-study at <https://github.com/yashpatel5400/fourier-cryo-splats>. It preserves failed development approaches and distinguishes them from completed outputs. The earlier reconstruction release v0.1.0 remains available with its conditional FSC results; it is not the uncertainty study's accuracy claim. No new conference submission or acceptance is implied by this format. Back-

ground PDFs are retained locally with source hashes rather than redistributed. Prematurely calibrated-looking maps can mislead structural interpretation; explicit assumptions and negative controls are central outputs of this work.

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Table 1. Fixed-pose ambient audit: median interval width relative to the no-data width across two density targets. All three methods after auditing use the same supported voxel class. The first column instead assumes the initial 33-dimensional dictionary is correct and is not an honest interval for that larger class. All entries are development results.

Dataset	Averaging width / field	Dictionary only	Same weights, audited	Enriched	Full matrix-free
10028	0.03	0.008	0.978	0.689	0.687
10028	0.07	0.007	0.510	0.053	0.050
10049	0.03	0.020	1.016	0.718	0.716
10049	0.07	0.020	0.636	0.075	0.073
10076	0.03	0.024	0.949	0.721	0.719
10076	0.07	0.023	0.462	0.072	0.072

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A. Additional Derivations

A.1. An identifiability lower bound

Proposition 1. *Let two allowed parameters induce $N(\mu_0, I)$ and $N(\mu_1, I)$, with targets t_0, t_1 . Any interval covering each target with probability at least $1 - \alpha$ has expected length under the first law at least*

$$|t_1 - t_0| [1 - 2\alpha - \text{TV}(P_0, P_1)]_+, \quad (13)$$

where $\text{TV}(P_0, P_1) = 2\Phi(\|\mu_1 - \mu_0\|/2) - 1$.

Proof. Under P_0 , the probability of containing t_1 is at least $1 - \alpha - \text{TV}(P_0, P_1)$ by the definition of total variation. The union bound gives simultaneous containment of both targets with probability at least the displayed positive part. The length on that event is at least their separation. The total variation formula follows by projecting onto the line between the Gaussian means. $\square$

This standard two-point argument also applies to different allowed nuisance settings. Exact null directions give a nonvanishing uncertainty floor even with perfect image prediction. It does not assert that a finite numerical search finds the largest possible separation.

A.2. Gaussian density-energy Gram matrix

For isotropic paired features, let

$$P_{jk} = \left(\frac{2\pi\sigma_j^2\sigma_k^2}{\sigma_j^2 + \sigma_k^2} \right)^{3/2} \exp \left[-\frac{\|\mu_j - \mu_k\|^2}{2(\sigma_j^2 + \sigma_k^2)} \right], \quad (14)$$

$$Q_{jk} = P_{jk} \text{ with } \mu_j - \mu_k \text{ replaced by } \mu_j + \mu_k.$$

Integrating products of Gaussians and applying Parseval gives real and imaginary coefficient blocks $2(P + Q)$ and $2(P - Q)$, with zero cross block. A zero-centered imaginary coefficient is exactly null. Numerically excluded eigenmodes are reported; exclusion changes the stated Gaussian subspace. In an ambient audit they remain available to the larger space rather than being silently assigned zero uncertainty.

A.3. Uniform nonlinear remainder

For a Gaussian of width σ at radial distance r , bounds on value, gradient and Hessian are

$$g_0(r) = e^{-r^2/(2\sigma^2)}, \quad (15)$$

$$g_1(r) = r g_0(r) / \sigma^2, \quad (16)$$

$$g_2(r) = (1 + r^2/\sigma^2) g_0(r) / \sigma^2. \quad (17)$$

On an interval of possible distances, g_0 is largest at the lower endpoint; g_1 and g_2 attain their largest value at the point

nearest σ . A rotation radius $a \leq \pi$ gives displacement at most $2\|k\| \sin(a/2)$, speed at most $a\|k\|$ and acceleration at most $a^2\|k\|$. A shift radius s gives phase speed at most $2\pi\|q\|s/\text{box}$.

The product rule bounds the second derivative by the sum of squared-speed times g_2 , acceleration times g_1 , twice speed times phase-speed times g_1 , and squared phase-speed times g_0 . Sum conjugate-pair bounds and apply the fixed CTF/noise scale. A coefficient-column Frobenius bound controls the operator term; the pilot-weighted bound controls its pilot term. Taylor’s integral formula gives Equation (11). Evaluating a Hessian only at the nominal pose would not establish this uniform result.

A.4. Simultaneous targets and sensitivity budgets

For K targets and parameter-budget choices fixed independently of inference noise, applying Theorem 1 with α/K gives simultaneous coverage by the union bound. Correlated Gaussian maxima can reduce the noise component if the joint weight covariance is accounted for. Selecting the most favorable point on a sensitivity surface after inspecting inference data requires such simultaneous control. A density radius supplied by simulation truth is an oracle control and cannot be reported as a practical calibration method.

B. Survey and Experimental Audit Trail

The repository’s living survey distinguishes density uncertainty, pose uncertainty, conformational variability/populations, map-processing confidence and atomic-model uncertainty. Its broad Europe PMC retrieval ledger contains 16,969 deduplicated records, mostly application papers; that count is not a count of reviewed methods. The curated list contains 86 candidate works at this draft checkpoint, with local full-text availability recorded separately. Recent items include LocScale-2.0, CryoDiff, CryoSplat, GEM, measurement-limited heterogeneity, population-counting work and the CAHRA dataset-construction preprint. Preprints and final publications are distinguished, and titles/versions are checked against primary sources.

The full experimental protocol draft records unimplemented comparisons and required failure controls. Fixed-pose resampling must be distinguished from rerunning alignment; a mathematical diagonal-Gaussian control must be distinguished from reproducing Ullrich et al.’s full software; cryoDRGN backprojection must be distinguished from training a neural model. Exposure-group splits do not undo the dependence introduced by full-data consensus poses. All development results, including the initially incomplete covariance-factor benchmark, remain identifiable by filename and status.